Synapses

1. **Concerning synapses:**
   a- axo-axonic is the most excitable.
   b- axo-dendritic is the most excitable
   c- axo-somatic is the least excitable.
   d- axo-dendritic is the least in number.

Answer: A

2. **concerning synapses:**
   a- electrical synapses are more common in CNS than chemical synapses
   b- chemical synapses act in both directions.
   c- Chemical synapses are characterized by presence of gap junctions between pre- and postsynaptic neurons.
   d- Chemical synapses show synaptic delay.

Answer: D

3. **Regarding the mechanism of synaptic transmission:**
   a- Ca^{2+} ions are released from the presynaptic knob.
   b- presynaptic knobs release only an excitatory transmitter.
   c- chemical transmitter causes the development of only excitatory post-synaptic potentials.
   d- excitatory transmitters open Na^{+} channels.

Answer: D

4. **Synaptic transmission is characterized by all of the following, EXCEPT:**
   a- impulses are transmitted from the pre to the postsynaptic neurons
   b- alkalosis depresses synaptic transmission.
   c- fatigue is due to exhaustion of neurotransmitter stores.
   d- repetitive stimulation leads to synaptic potentiation.

Answer: B

5. **Released transmitter is inactivated by all of the following except:**
   a- active re-uptake.
   b- inactivation by enzymes.
   c- diffuse away in ECF.
   d- phagocytosed by glial cells.

Answer: D

6. **Regarding synaptic potentials:**
   a. presynaptic inhibition causes hyperpolarization of post-synaptic neuron.
   b. during IPSP a neuron can not discharge an action potential.

c. Post-tetanic facilitation is due to increased secretion of chemical transmitter due to stimulation by a 3rd excitatory presynaptic neuron.
d. EPSP can be caused by glycine.

Answer: B

7. Excitatory postsynaptic potential:

a. can not be summated.
b. is a state of local depolarization.
c. is due to opening of voltage gated Na^+ channels.
d. is due to opening of chemical gated K^+ channels.

Answer: B

8. Excitatory postsynaptic potential:

a- is due to increase permeability of the postsynaptic membrane to Na^+
b- is a state of complete depolarization.
c- can produce an action potential without summation.
d- of dendrites can reach to an action potential.

Answer: A

9. EPSPs:

a- result from the release of GABA from presynaptic neurons.
b- can be temporally summated.
c- result from increased permeability of the postsynaptic membrane to K^+
d- obeys the all-or-none rule.

Answer: B

10. Excitatory postsynaptic potential:

a- is not propagated.
b- shows a reversal of polarity.
c- lasts about 2 seconds.
d- may reach an action potential at the soma without summation.

Answer: A

11. An inhibitory neurotransmitter increases the permeability of the postsynaptic neuron to:

a- K^+
b- Na^+
c- Ca^{2+}
d- HCO_3^-

Answer: A

12. Inhibitory postsynaptic potentials:

a- is due to presynaptic inhibition.
b- causes a state of hyperpolarization.
c- is associated with an influx of Na+ ions.

d- cannot be summated.

Answer: B

13. IPSP:

a- is a state of depolarization of the postsynaptic membrane.
b- is produced by binding of acetylcholine to its receptor.
c- results from the inflow of Cl^- ions and outflow of K^+ ions from the postsynaptic membrane.
d- results from the release of catecholamines from the presynaptic knob.

Answer: C

14.Inhibitory post synaptic potential:

a- involves increased K+ efflux.
b- involves increased Na+ efflux.
c- is a state of partial depolarization.
d- increases excitability of the postsynaptic membrane.

Answer: A

15.Regarding presynaptic inhibition:

a- the postsynaptic neuron membrane is hyperpolarized.
b- there is decreased release of excitatory chemical transmitters from the presynaptic neuron.
c- there is increased Ca^{2+} influx into presynaptic neuron.
d- there is closure of K^+ channels of the presynaptic neuron.

Answer: B

16. Inhibitory presynaptic potential:

a- needs an inhibitory third neuron.
b- needs excitatory third neuron.
c- leads to depolarization of the presynaptic neuron.
d- directly inhibits the postsynaptic neuron.

Answer: A

17.Regarding synaptic transmission:

a- is depressed by alkalosis.
b- is increased by acidosis.
c- repetitive stimulation of a single presynaptic neuron leads to the development of an action potential in the postsynaptic neuron
d- is increased by hypoxia.

Answer: C

18.Synaptic transmission:

a- is increased by alkalosis.
b- shows a delay of about 0.05 ms.

c- is increased by acidosis.
d- decreased during tetanus disease

Answer: A

19. Synaptic transmission is facilitated by all of the followings, EXCEPT:

a- coffee and tea.
b- alkalosis.
c- strychnine.
d- hypoxia.

Answer: D

20.Regarding synaptic transmission, it is decreased by:

a- alkalosis.
b- hyperventilation.
c- increased secretion of GABA
d- theobromine.

Answer: C

21.If the central delay in one reflex is 4 msec., then the:

a- number of synapses in this reflex is two.
b- number of synapses in this reflex is four.
c- number of synapses in this reflex is eight.
d- it is a monosynaptic reflex.

Answer: C

22.Synaptic plasticity shows all of the following, EXCEPT:

a- ability to change the function of the synapse according to the demand.
b- play an important role in memory.
c- may lead only to long term potentiation.
d- explain the process of habituation.

Answer: C

23.Sensitization of synapses:

a- is due to repetitive benign stimuli which is associated with noxious stimuli.
b- is the base of long-term memory.
c- is explained by postsynaptic facilitation.
d- causes hyperpolarization of the postsynaptic neuron.

Answer: A

24. Glycine:

a- acts mainly as inhibitory chemical transmitter.
b- action is potentiated by strychnine.
c- causes depolarization of the postsynaptic membrane.
d- is responsible for the process of sensitization

Answer: A

25. Regarding rapidly acting transmitters, they are:

a- formed in the soma.
b- formed in small amounts.
c- very potent.
d- responsible for rapid neurological response.

Answer: D

26. Concerning the slowly acting neurotransmitters, they are:

a- formed in the synaptic knob.
b- formed in huge amounts.
c- stored in clear vesicles in synaptic knobs
d- very potent

Answer: D

27. Regarding GABA, all of the following statements are true, EXCEPT:

a- it is an excitatory neurotransmitter in the brain.
b- it produces an IPSP.
c- it decreases Ca^{2+} influx into the pre-synaptic neuron.
d- it increases Cl^- influx and K^+ efflux in the post-synaptic neuron.

Answer: A

28. NMDA receptors:

a- are stimulated by acetylcholine.
b- are both ligand and voltage-gated calcium channels.
c- are blocked by Ca^{2+} at normal membrane potential
d- are inhibited by consequent activation of AMPA receptors.

Answer: B

29. Renshaw cell:

a- shows the phenomenon of lateral inhibition.
b- is activated by glycine.
c- is present in dorsal horn of the spinal cord.
d- are excitatory neurons.

Answer: A

30. All of the following about synaptic transmission is true EXCEPT:

a- It is unidirectional from presynaptic to postsynaptic neuron.
b- It undergoes fatigue by repeated stimulation.
c- It shows a delay by about 5 msec at each synapse.
d- It is depressed by anesthetics and hypnotics.

Answer: C

31. Which of the following is characteristic of events occurring at excitatory synapses?

a) synaptic vesicles bind to postsynaptic membrane.
b) voltage-gated potassium channels are closed.

c) efflux of calcium from the presynaptic terminal.
d) ligand-gated sodium channels are opened

Answer: D

32.regarding summation:

a) Spatial summation is more common than temporal summation.
b) Temporal summation requires discharge from multiple presynaptic neurons.
c) Spatial summation requires repetitive discharge from single presynaptic neuron.
d) summation is rarely required in synapses.

Answer: A

33.which of the following is an inhibitory neurotransmitter:

a. Glutamate.
b. Glycine.
c. Aspartate.
d. Adrenaline.

Answer: B

Somatosensory function

34. Pacinian corpuscles are:

a- pain receptors.
b- rapidly adapting pressure receptors.
c- slowly adapting touch receptors.
d- innervated by C nerve fibers.

Answer: B

35. Pacinian corpuscles are:

a- A type of temperature receptors.
b- usually innervated by Aδ nerve fibers.
c- rapidly adapting receptors.
d- pain receptors.

Answer: C

36.Rods and cones of the retina are a type of:

a- Mechanoreceptors.
b- Thermoreceptors.
c- Nociceptors.
d- Photoreceptors.

Answer: D

37.Touch receptors are a type of:

a- Mechanoreceptors.

b- Thermoreceptors.
c- Nociceptors.
d- Photoreceptors.
Answer: A

38. Taste buds contain receptor cells that considered a type of:

a- Mechanoreceptors.
b- Thermoreceptors.
c- Nociceptors.
d- chemoreceptors.
Answer: D

39. Which of the following receptors are NOT found in the skin:

a. Mechanoreceptors.
b. Thermoreceptors.
c. Nociceptors.
d. Photoreceptors.
Answer: D

40. Specificity:

a- explains the process of adaptation.
b- means that each receptor is most sensitive to a specific form of energy called the adequate stimulus.
c- is a state of partial depolarization.
d- is due to high intensity of the stimulus.
Answer: B

41. Generator potential:

a- is a state of partial depolarization.
b- duration is shorter than that of the action potential.
c- is accompanied with absolute refractory period.
d- can not be summated.
Answer: A

42. Receptor potential:

a- is an action potential.
b- is due to opening of Na^+ channels.
c- duration is shorter than that of the action potential.
d- brings the membrane potential away from the firing level.
Answer: B

43. The receptor potential:

a- is conducted actively to the 1st node of Ranvier.
b- can be summated.
c- is accompanied with absolute refractory period.
d- has the same duration like that of the action potential.
Answer: B

44. Fast adaptation receptors include:

a- pain receptors.
b- auditory receptors.
c- Pacinian corpuscles.
d- arterial baroreceptors.
Answer: C

45. Discrimination of the intensity of stimuli results from the:
a- presence of adequate stimulus for each receptor.
b- presence of a specific pathway for each sensation.
c- specific center for each sensation.
d- Weber-Feschner's law or Powers low.
Answer: D

46. The size of representation of various body parts in the primary somatic sensory cortex is dependent on:
a- relative size of the body parts.
b- density of the receptors in that organ.
c- size of the muscles in that body part.
d- conduction velocity of the afferent fibers
Answer: B

47. Fine touch sensations:
a- are not sharply localized.
b- need a relatively high threshold stimulus.
c- are transmitted via the dorsal column.
d- are perceived by free nerve endings.
Answer: C

48. Fine touch sensation:
a- is perceived by Merkel's discs.
b- are conducted by C fibers.
c- is finally relayed in the thalamus.
d- is perceived more on the back of the shoulder.
Answer: A

49.Two-point discrimination is more accurate:
a- at the back.
b- the more the convergence of afferents.
c- the more the number of receptors.
d- the less the cortical representation.
Answer: C

50. Action potentials recorded after stimulation of thermal receptors show that:
a- The maximal rate of discharge of action potentials in the cold nerve fibers is at 25°C.
b- action potentials are recorded in warm nerve fibers between 10-35°C.

c- action potentials are recorded in cold pain fibers at 0°C.
d- the maximum rate of discharge from warm pain and cold pain fibers occurs at 40°C.
Answer: A

51. The distance by which two touch stimuli must be separated to be perceived as two separate stimuli is greatest on:
a- the lips.
b- the palm of the hand.
c- the back of the scapula.
d- the tips of the finger.
Answer: C

52. Vibration sense:
a- is sensed by free nerve endings.
b- depends on past cortical experience.
c- is carried on gracile and cuneate tracts.
d- is carried on the spinothalamic tract.
Answer: C

Pain sensation

53. The fast pain sensation that accompanies a painful trauma is carried to C.N.S. by fibers that belong to:
a- A alpha fibers.
b- A delta fibers.
c- C fibers.
d- B fibers.
Answer: B

54. Pain:
a- receptors are stimulated by a rise in bradykinin concentration.
b- threshold differs from one person to another.
c- sensation is enhanced by endorphins.
d- sensation is enhanced by GABA released by Aβ fibers.
Answer: A

55. The pricking pain:
a- terminate in the non-specific thalamic nuclei.
b- is lost by compression of the nerve.
c- is blocked by local anesthetics better than slow pain.
d- is a severe dull aching pain.
Answer: B

56. Slow pain:

a- is carried along type A delta fibers.
b- pathway activates the non-specific thalamic nuclei.
c- is perceived mainly in sensory area I.
d- lasts for a short time.

Answer: B

57. Cutaneous pain:

a- receptors adapt to stimulation more rapidly than touch receptors.
b- can be caused by the excitation of receptors by K^+ ions released in injured tissues.
c- can be caused by over-stimulation of touch receptors.
d- is carried into the spinal cord by a long A-beta fibers.

Answer: B

58. Cutaneous pain:

a- threshold is the same for males and females.
b- is carried only on myelinated nerve fibers.
c- is carried on the spinothalamic tract of the same side.
d- may refer to the viscera.

Answer: A

59. Deep pain:

a- is due to stimulation of pain receptors in the skin.
b- is pricking in character.
c- is poorly localized.
d- is carried along A-delta fibers.

Answer: C

60. Concerning visceral pain, the following are true statements, EXCEPT:

a- may cause reflex vomiting.
b- may cause contraction of overlying skeletal muscle.
c- is well localized.
d- may cause a change in arterial blood pressure.

Answer: C

61. All the following may give rise to visceral pain, EXCEPT:

a- distension of hallow viscus.
b- ischemia.
c- spasm of hallow viscus.
d- incision with scalpel in the gut wall.

Answer: D

62. Visceral pain:

a- is a localized pain.
b- is accompanied by a spasm of its overlying muscle.
c- can be elicited by sharp cutting of viscera.
d- is carried along Gracile and cuneate tracts.

Answer: B

63. Visceral pain results from all of the following, EXCEPT:

a- ischemia which releases bradycardia.
b- Escape of HCl through a ruptured duodenal ulcer.
c- overdistension of hollow organ.
d- cutting through the viscera.

Answer: D

64. Gallbladder pain is referred to the:

a- umbilicus.
b- tip of the right scapula.
c- left shoulder.
d- left lumbar region.

Answer: B

65. Cardiac pain is referred to the:

a- umbilicus.
b- tip of the right scapula.
c- left shoulder.
d- left lumbar region.

Answer: C

66. The analgesia system includes:

a- Raphe magnus nucleus in the posterior horn of the spinal cord.
b- the periaqueductal area in the mid-brain.
c- the sensory area I.
d- supra-optic nuclei of the hypothalamus.

Answer: B

67. Serotonin is used as neurotransmitter in pain suppression by neuron in the following region:

a- raphe magnus nucleus.
b- periaqueductal gray matter.
c- post-central gyrus.
d- nucleus cuneatus.

Answer: A

68. An interneuron in which region uses enkephalin to inhibit pain transmission:

a- post-central gyrus.
b- raphe magnus nucleus.
c- dorsal horn of the spinal cord.
d- periaqueductal gray matter.

Answer: C

69. The spinothalamic system transmits the following sensations:

a- Proprioceptive sensations.
b- vibration sensation.
c- pain and temperature.
d- two-point discrimination

Answer: C

70. The main final destination of signals that carry kinesthetic sensations is the:

a- post-central gyrus of the cerebral cortex.
b- ventrolateral nuclei of the thalamus.
c- cerebellar cortex.
d- caudate nucleus.

Answer: A

71. Regarding secondary hyperalgesia:

a- pain threshold is low.
b- may be due to sun burn.
c- skin is reddish in colour.
d- In the healthy skin around the red area.

Answer: D

72. Concerning primary hyperalgesia:

a- pain threshold is high.
b- may be due to sun burn.
c- skin is normal.
d- may be caused by thalamic syndrome.

Answer: B

73. Posterior column damage of the spinal cord may impair the following, EXCEPT:

a- the ability to stand steadily with the eyes closed.
b- pain sensation.
c- vibration sensation.
d- kinesthetic sensations.

Answer: B

74. Which of the following is related to the analgesia system and present in the spinal cord:

a) Periventricular grey area
b) Periaqueductal grey area
c- Raphe magnus nucleus
d- pain inhibitory complex

Answer: D

75. the center of stereognosis is:

a- Somatic sensory area I.
b- Sensory area II.
c- Sensory association area.
d- Parieto-occipito-temporal association area.

Answer: C

76. initiation of fine movements is the function of:

a. Motor area 4
b. supplementary motor area
C. premotor area
D. superior temporal area.

Answer: A

Spinal reflexes

77. The stretch reflex:

a) is a polysynaptic reflex.
b) is a monosynaptic reflex.
c) receptor is the Pacinian corpuscle.
d) has a long central delay.

Answer: B

78. Regarding the muscle spindle:

a) the primary afferent fibres respond to the static length of muscle as well as to its rate of change in length.
b) stretching the extrafusal fibres causes the intrafusal muscle fibers to be stretched.
c) they are excited during overstretch of the muscle.
d) they are in series with the extrafusal muscle fibers.

Answer: B

79. All are true regarding the stretch reflex, EXCEPT:

a) can be elicited by a passive stretch of the muscle.
b) depends on the muscle spindle which is in parallel with the extrafusal

muscle fibers.

c) it is a polysynaptic reflex.
d) stretch of a muscle is followed by its contraction.

Answer: C

80. The gamma efferent system:

a) Modifies the muscle spindle sensitivity.
b) supplies the central zone of intrafusal muscle fibres.
c) is inhibited by the vestibulospinal tract.
d) Supplies the extrafusal muscle fibers.

Answer: A

81. The primary endings innervating the muscle spindle:

a) only innervates the nuclear chain fibres.
b) innervate both nuclear chain and nuclear bag fibres.
c) are a slowly conducting nerves having a diameter of 4 microns.
d) discharge during the static response only.

Answer: B

82. The nuclear bag intrafusal fibers that signal the rate of changes in muscle length is innervated by:

a) corticospinal neurons.
b) alpha motor neurons.
c) dynamic gamma motor neurons.
d) static gamma motor neurons.

Answer: C

83. Gamma motor nerve fibers:

a) cause the contraction of the peripheral part of the intrafusal fibers.
b) cause contraction of the middle portions of the muscle spindle.
c) cause relaxation of the intrafusal fibres.
d) Thickness is about 16 μ.

Answer: A

84.The secondary endings innervating the muscle spindle:

a. only innervates the nuclear chain fibers.
b. only innervates the nuclear bag fibers.
c. are a rapidly conducting nerves having a diameter of 16 microns.
d. discharge during the dynamic response only.

Answer: A

85. Muscle tone, all are true, EXCEPT:

a) is defined as resistance offered by muscle to passive stretch.
b) is maintained by gamma efferent discharges.
c) maintains body posture against effect of gravity.

d) is decreased in UMNL.

Answer: D

86. Regarding the muscle tone, all are true, EXCEPT:

a) the primary and secondary afferents are both stimulated when the spindle is stretched.
b) the gamma efferent discharges are firing independently of the descending tracts from areas in the brain.
c) hypertonia may occur during hyperthyroidism.
d) hypotonia commonly occurs as a result of spinal motor neuron damage.

Answer: B

87. All of the followings are inhibitory supraspinal centers to the stretch reflex, EXCEPT:

a. premotor area 6.
b. Neocerebellum.
c. lentiform nucleus of the basal ganglia
d. Red nucleus.

Answer: B

88. All of the followings are facilitatory supraspinal centers to the stretch reflex, EXCEPT:

a. Pontine reticular formation.
b. Neocerebellum.
c. vestibular nucleus
d. premotor area 6

Answer: D

89. Hyporeflexia is present in:

a) Parkinsonism.
b) LMNL.
c) hyperthyroidism.
d) UMNL.

Answer: B

90. Tendon jerks are exaggerated in:

a) anterior horn cell disease.
b) dorsal root lesions.
c) Chorea.
d) internal capsule lesion.

Answer: D

91. Golgi tendon organ:

a) has a threshold of stimulation equal to the muscle spindle.
b) lies in parallel with the extra-fusal muscle fibers.
c) when stimulated reflexely cause the contraction of the muscle in the

tendon of which it is situated.

d) when stimulated, reflexely relaxes the muscle in the tendon of which it is situated.

Answer: D

92. The only reflex which doesn't obey the phenomenon of reciprocal innervation is:

a) Stretch reflex.
b) Inverse stretch reflex.
c) Flexion withdrawal reflex.
d) Positive supporting reaction.

Answer: D

93. The ankle jerk is exaggerated:

a) LMNL.
b) in Parkinsonism.
c) in neocerebellar disease.
d) on the right side after damage in the left internal capsule.

Answer: D

94. Muscle tone is reduced in the following conditions, EXCEPT:

a) lower motor neuron lesion.
b) upper motor neuron lesion.
c) cerebellar lesions.
d) toxicity with curare-like drugs.

Answer: B

95. The inverse stretch reflex:

a) depends on a receptor which is situated in the middle of the muscle.
b) is elicited when the muscle is subjected to excessive stretch.
c) leads to increased force of muscle contraction.
d) is a monosynaptic reflex.

Answer: B

96. The withdrawal reflex:

a) is monosynaptic.
b) is polysynaptic.
c) is more rapid than the stretch reflex.
d) disappears in UMNL.

Answer: B

97. the receptor of the inverse stretch reflex is:

a. muscle spindle
b. Golgi tendon organ
c. Pacinian corpuscle.
d. Free nerve endings

Answer: B

98. The extrafusal muscle fibers are innervated by :

a. lateral spinothalamic neurons.
b. alpha motor neurons.
c. dynamic gamma motor neurons.
d. static gamma motor neurons.

Answer: B

99. The nuclear chain intrafusal fibers are innervated by :

a. corticospinal neurons.
b. alpha motor neurons.
c. dynamic gamma motor neurons.
d. static gamma motor neurons.

Answer: D

100. Motor neurons in the spinal cord may be the site of convergence of signals arising from all the following sites, EXCEPT:

a) cerebral cortex.
b) cerebellum.
c) reticular formation.
d) cells in the dorsal root ganglia.

Answer: B

101. Regarding the crossed extensor reflex, all are true, EXCEPT:

a) is an antigravity reflex.
b) is a polysynaptic spinal reflex.
c) has a shorter latency than the stretch reflex.
d) never appear after spinal cord transection.

Answer: C

102. Concerning the flexor withdrawal reflexes:

a) they involve reciprocal innervation.
b) their central delay equals 0.5 msec.
c) it is a monosynaptic reflex.
d) it provides an example of a deep reflex.

Answer: A

103. Regarding the muscle spindle:

a) Has an afferent connection belonging to Ia fibers.
b) Has an efferent connection belonging to type II fibers.
c) Contains the extrafusal muscle fibers.
d) Responds when the muscle is overstretched.

Answer: A

104. The afferent from the Golgi tendon organs belongs to:

a) Group Ia fibers b) group Ib fibers c) group II fibers d) group III fibers

Answer: B

105. Which of the following receptors receives an efferent:

a) Pacinian corpuscle
b) Muscle spindle
c) Golgi tendon organ
d) Pain receptors

Answer: B

brain higher centers

106. Degeneration of dopamine-containing neurons in the substantia nigra produces:

a) Choreiform movements.
b) hemiballismus.
c) suppression of activity in the subthalamic nucleus.
d) parkinsonism.

Answer: D

107. Maintenance of equilibrium depends on information reaching C.N.S. from the following structures, EXCEPT:

a) proprioceptors in muscle and joints.
b) cochlear nerve.
c) retina
d) vestibular apparatus.

Answer: B

108. A lesion of Broca's area produces:

a) apraxia.
b) motor aphasia.
c) agraphia.
d) word deafness.

Answer: B

109. The ascending reticular formation. All are true, EXCEPT:

a) is activated by collateral branches of sensory neurons transmitting impulses to the thalamus.
b) sends impulses to most parts of the cerebral cortex.
c) when stimulated electrically tends to increase alertness.
d) it relays mainly in specific thalamic nuclei.

Answer: D

110. The otolith organs are receptors for:

a) linear acceleration.
b) Angular acceleration.
c) proprioception.

d) vibration.

Answer: A

111. All are true about the vestibular apparatus, EXCEPT:

a) bony labyrinth contains 3 SCCs.
b) helps to maintain equilibrium.
c) is stimulated only during linear acceleration.
d) is stimulated during linear and angular acceleration.

Answer: C

112. During rotation in one plane, all are true, EXCEPT:

a) the hair cells of the cristae are stimulated.
b) cristae of the horizontal canals are stimulated when it bends towards the utricle.
c) cristae of vertical canals is stimulated when it bends away from the utricle.
d) all SCCs are stimulated.

Answer: D

113. All are true about utricles and saccules, EXCEPT:

a) have their cells arranged in different planes.
b) are stimulated by changing the position of the head.
c) have macula as their receptors.
d) respond to angular acceleration

Answer: D

114. Parkinson's disease:

a) is caused by a lesion in the substantia nigra.
b) is caused by lesion in the posteroventral nucleus of the thalamus.
c) is accompanied by kinetic tremors.
d) is accompanied by clasp-knife rigidity.

Answer: A

115. All of the following are features of Parkinsonism, EXCEPT:

a) a tremor at rest.
b) a deficiency of dopamine in the substantia nigra.
c) a clasp-knife type of spastic rigidity.
d) difficulty in initiating the voluntary movements.

Answer: C

116. Parkinson's disease is characterized by:

a) tremors made worse by movements.
b) paralysis of one side of the body.
c) increase of dopamine in the substantia nigra.

d) lead-pipe rigidity.

Answer: D

117. Parkinsonism results in all, EXCEPT:

a) akinesia.
b) kinetic tremors.
c) mask face.
d) hypertonia.

Answer: B

118. Parkinsonism is associated with:

a) kinetic tremors.
b) adidochokinesia.
c) rebound phenomenon.
d) shuffling gait.

Answer: D

119. Chorea:

a) is due to lesion in the substantia nigra.
b) may be due to lesion in the internal capsule.
c) is associated with hypotonia.
d) is associated with exaggerated tendon jerk.

Answer: C

120. The functions of the cerebellum include all of the following except:

a) stop the movement at the precise intended point.
b) prevents oscillation of movements.
c) planning and timing function.
d) Initiation of fine movements.

Answer: D

121. Regarding the cerebellum:

a) has no role in maintenance of balance.
b) neocerebellum inhibits muscle tone.
c) poleocerebellum facilitates muscle tone.
d) cerebellum has a predictive function during running.

Answer: D

122. Neocerebellar syndrome in humans is characterized by all, EXCEPT:

a) hypotonia.
b) pendular knee jerk.
c) disturbance of balance.

d) static tremors.

Answer: D

123. Adiadochokinesia:

a) is failure of the patient to perform rapid coordinate movements.
b) is due to lesion in caudate nucleus.
c) is due to lesion in subthalamus.
d) is due to lesion in red nucleus.

Answer: A

124. The following are neurological features of cerebellar lesion, EXCEPT:

a) abnormal finger-nose test.
b) rebound phenomenon.
c) dysdiadochokinesia.
d) shuffling gate.

Answer: D

125. All about kinetic tremors is true, EXCEPT:

a) occurs during rest.
b) is due to neocerebellar lesion.
c) is accompanied with hypotonia.
d) disappear during sleep.

Answer: A

126. Neocerebellum syndrome is associated with all, EXCEPT:

a) ataxia.
b) hypertonia.
c) inability to do complex movements.
d) rebound phenomenon.

Answer: B

127. Neocerebellum disease is associated with:

a) shuffling gait.
b) drunken gait.
c) stamping gait.
d) circumduction gate.

Answer: B

128. A cerebellar lesion is most likely associated with severe loss of equilibrium when the lesion involves:

a) the neocerebellum.
b) a small portion of the right hemisphere.

c) a small portion of the left hemisphere.
d) the flocculonodular lobe.

Answer: D

129. The following are neurological features of cerebellar lesion, EXCEPT:

a) dysmetria.
b) lead-pipe rigidity.
c) dysdiadochokinesia.
d) rebound phenomenon.

Answer: B

130. Delta waves can be obtained in EEG of a:

a-normal wakeful person.
b-during REM sleep.
c-person who sees a bright light.
d-non-REM sleep.

Answer: D

131. EEG record of relaxed subject, who closes his eyes include the following wave:

a-alpha.
b-beta.
c-theta.
d-delta.

Answer: A

132. If the subject think or open his eyes, EEG record wave will be:

a-alpha.
b-beta.
c-theta.
d-Delta

Answer: B

133. All is true about alpha waves of EEG, EXCEPT:

a-disappear when a person solves a problem.
b-replaced by beta waves during mental activity.
c-has a frequency of 30 c/sec.
d-has an amplitude of 50 μv.

Answer: C

134. Which of the following structures is maximally sensitive

to linear movement of head in the horizontal plane?

a) macula of saccule.
b) macula of utricle.
c) crista ampullaris of semi-circular canals.
d) proprioceptors in the neck.
Answer: B

135. Slow wave sleep is associated with:

a) increased respiratory rate.
b) increase muscle tone.
c) decrease ABP.
d) dream which can be remembered.

Answer: C

136. A patient who suffers from hemiplegia is able to write his thought to his family but is unable to express these ideas as spoken words is affected by:

a) visual agnosia.
b) sensory aphasia.
c) motor aphasia.
d) apraxia.

Answer: C

137. Motor apraxia:

a) is due to lesion in primary motor area 4.
b) is due to lesion in sensory area.
c) is accompanied with muscle paralysis.
d) is characterized by loss of memory of voluntary movements.

Answer: D

138. During REM sleep:

a) muscle tone throughout the body is increased.
b) Beta waves are observed in EEG.
c) individuals rarely awake spontaneously at its end.
d) is a dreamless sleep.
Answer: B

139. The portion of the basal ganglia concerned with control of cognitive motor activity (memory guided) involves the following:

a) Putamen.
b) caudate nucleus.
c) globus pallidus.

d) subthalamus.

Answer: B

140. Horizontal rotation of the head is likely to stimulate:

a) macula of the utricle.
b) crista ampullaris of the posterior semicircular canal.
c) crista ampullaris of the anterior semicircular canal.
d) crista ampullaris of the horizontal semicircular canal.

Answer: D

141. Concerning the functions of the cerebellum the following statements are true, EXCEPT:

a) the cerebellum co-ordinates the muscle contractions required to execute a movement.
b) the cerebellum directly stimulates motor neurons required to make a movement.
c) the cerebellum receives feedback from muscles that execute the movement.
d) the cerebellum is involved in the planning of a movement.

Answer: B

142. The part of the basal ganglia that release an excitatory transmitter (glutamate) is:

a) Caudate nucleus.
b) Putamen.
c) Globus pallidus.
d) Subthalamic nucleus.

Answer: D

143. parkinsonism results from damage of which of the following:

a) substantial nigra.
b) thalamus.
c) Subthalamus.
d) Cerebellum.

Answer: A

motor, sensory areas and motor tracts

144. The larger area of representation in the motor cortex includes muscles:

a) that extend the shoulder.

b) that flex the knee.
c) of the trunk.
d) of vocalization.

145. Regarding the primary motor cortex (area 4), all are true, EXCEPT:
a) is responsible for fine delicate movements.
b) parts of the body are represented in an area which is directly proportional in size to the complexity of movement carried by that part.
c) sends facilitatory impulses to gamma motor neurons.
d) is inhibitory to muscle spindles.

146. The cortical motor area 4 on the left side of the body:
a) there is a map representing body parts on the right side.
b) the body is represented in an erect (upright) pattern.
c) controls motor activity of the left side of the body.
d) the area representing a particular body part is directly proportional in size to the size of that part or organ.

147. The pyramidal tract:
a) arises only from area 4.
b) supplies only the same side of the body.
c) most of the fibres do not cross.
d) responsible for fine movements.

148. Most of fibres of the pyramidal tract have direct synapse with:
a) anterior alpha motor neurons.
b) gamma motor neurons.
c) interneurons of the spinal cord.
d) dorsal horn cells of spinal cord.

149. The pyramidal tract originates from:
a) all areas of cerebral cortex including parietal, frontal, temporal and occipital lobes.
b) motor, premotor areas as well from somatic sensory cortex.
c) area 4 only.
d) motor area 4 and premotor area 6 only.

150. In controlling the fine movement of the hands, the corticospinal tract terminates directly on:
a) Renshaw cells .
b) spinal cord interneurons .

c) spinal alpha motor neurons .
d) posterior horn cells .

151. Babinski sign is present in all, EXCEPT:

a) pyramidal lesion.
b) extrapyramidal lesion.
c) cerebellar lesion.
d) deep sleep.

152. Babinski sign is present in:

a) pyramidal tract lesion.
b) chorea.
c) children over 5 years.
d) lower motor neuron lesion

153. Babiniski sign is characterized by:

a) plantar flexion of the great toe and fanning out of the other toes.
b) dorsiflexion of the great toe and fanning out of the other toes.
c) spontaneous involuntary flexor movements of the stimulated limb.
d) dissociated sensory loss.

Special senses

154. The fovea:

a) is more vascular than the periphery of the retina
b) is composed of cones only
c) contains a greater proportion of ganglion cells
d) contains a greater proportion of rods than the periphery of the retina

Answer: B

155. Aqueous humour:

a) pressure is about 90 mmHg.
b) is easily absorbed when the pupil is widely dilated.
c) formation is dependent on carbonic anhydrase enzyme.
d) is absorbed by the choroid's plexus.

Answer: C

156. The primary visual cortex:

a) is located in the temporal lobe.
b) lies adjacent to the calcarine fissure of the occipital cortex.
c) contains areas for form, colour and details of objects.

d) is called V-2.

Answer: B

157. The neural elements responsible for the accommodation of the lens when the eyes are focused on a near object include:

a) the parasympathetic fibers in the trochlear nerve.

b) the parasympathetic fibers in the oculomotor nerve.

c) sympathetic nerve fibers in oculomotor nerve.

d) sympathetic nerve fibers in abducent nerve.

Answer: B

158. The condition in which the lens of the eye becomes totally unaccommodating in elderly persons is:

a) myopia.

b) emmetropia.

c) presbyopia.

d) hypermetropia.

Answer: C

159. The aqueous humor is secreted by:

a) iris.

b) cornea.

c) lens.

d) ciliary processes.

Answer: D

160. Which of the following photoreceptors respond to the longest wavelength of light?

a) rods.

b) green cones.

c) blue cones.

d) red cones.

Answer: D

161. Regarding colour vision, which of the following statements is true?

a) the simulation ratio of the three types of cones allows specific colour perception.

b) yellow colour is perceived when green and blue cones are stimulated equally.

c) when there is no stimulation of red, blue or green cones, there is the sensation of seeing white.

c- The curvature of the cornea increases.
d- Chromatic and spherical aberration is decreased.

Answer: C

168. When light is shone into one eye, the pupil:

a- in that eye constricts and in the opposite pupil dilates.
b- constricts even though its optic nerve has been cut.
c- responds due to parasympathetic nerve activity.
d- responds even after local application of atropine.

Answer: C

169. Presbyopia is caused by:

a- weakness of the ciliary muscle.
b- loss of lens elasticity.
c- degeneration of the retina.
d- clouding of the cornea.

Answer: B

170. Activation of transducin by light activates an enzyme that:

a- activates adenyl cyclase.
b- hydrolyzes cGMP.
c- depolarizes the receptor membrane.
d- releases calcium from intracellular stores.

Answer: B

171. The Argyll Roberston pupil:

a- do not constrict because the parasympathetic fibers to the eye are destroyed.
b- do not constrict because the sympathetic fibers to the eye are destroyed.
c- constrict during accommodation when looking at near object.
d- constrict when light is shone on the eye.

Answer: C

172. The fovea of the eye:

a- contains only rods.
b- is situated over the exit of the optic nerve.
c- is the region of highest visual acuity.
d- is the thickest part of the retina.

Answer: C

173. Which of the following parts of the eye has the highest concentration of rods?

a- fovea.

b- peripheral part of retina.
c- iris.
d- optic disc.

Answer: B

174. Visual accommodation involves:

a- increased intraocular pressure.
b- increased tension on the lens ligaments.
c- contraction of the ciliary muscle.
d- dilatation of the pupil.

Answer: C

175. In response to light, the following changes occur in the rods, EXCEPT:

a- decreased cGMP.
b- activation of transducin.
c- closure of Na^+ channels in the outer segment.
d- increased release of synaptic transmitter.

Answer: D

176. Abnormal colour vision is caused by:

a- recessive gene on the Y chromosome.
b- dominant gene on the Y chromosome.
c- recessive gene on the X chromosome.
d- recessive gene on chromosome 7.

Answer: C

177. The following event occurs in photoreceptors during phototransduction in response to light:

a) phosphodiesterase activity decreases.
b) hydrolysis of cGMP increases.
c) neurotransmitter release increases.
d) number of open voltage-gated calcium channels increases.

Answer: B

178. The following are functions of the middle ear, EXCEPT:

a) the Eustachian tube equalizes pressure on both sides of the oval window.
b) impedance matching to transfer effectively sounds from the air-filled middle ear to the fluid filled cochlea

c) the constant contraction of the middle ear muscles, keeps the tympanic membrane optimally tense for vibration by sounds.
d) attenuation of sound by the reflex contraction of middle ear muscles.

Answer: A

179. Which of the following is the middle ear ossicles that is attached to the tympanic membrane?

a) malleus
b) incus
c) stapes
d) modiolus

Answer: A

180. Tympanic reflex:

a) protects the inner ear from the vibrations of high-frequency sounds.
b) increases the rigidity of the ossicular system, thereby reducing conduction of high intensity sounds.
c) occurs following a latent period of 4 to 8 seconds after a loud sound.
d) can increase the intensity of low-frequency sound transmission by 30 to 40 decibels.

Answer: B

181. Regarding the basilar membrane, the following is true:

a) it separates the scala media from the scala vestibule.
b) its fibers decrease in length from the base of the cochlea to the apex of the cochlea.
c) its fibers increase in diameter from the base to the apex of the cochlea.
d) it vibrates best at high frequency near the base of the cochlea whereas it vibrates at low frequency at the apex of the cochlea.

Answer: D

182. The organ of Corti is contained within:

a) saccule.
b) scala media.
c) scala tympani.
d) scala vestibule.

Answer: B

183. Which of the following muscles is contracted as part of the papillary light reflex?

a) dilator pupillae muscle.
b) constrictor pupillae muscle.

c) ciliary muscle.
d) medial rectus.
Answer: B

184. Concerning the discrimitation of sound loudness, the following is true:
a) amplitude of vibration of the basilar membrane increases with louder sounds
b) with loud sounds hair cells excite nerve endings at a diminished rate.
c) few outer hair cells are stimulated with loud sounds.
d) few inner hair cells are stimulated with loud sounds.
Answer: B

185. In the middle ear:
a) air pressure is normally subatomspheric.
b) destruction of auditory ossicles abolishes hearing.
c) the round window moves reciprocally with the oval window.
d) paralysis of the middle ear muscles abolishes hearing.
Answer: C

186. The tympanic membrane:
a- stops vibration almost immediately after the sound stops.
b- transmits sound more effectively when the muscles of the middle ear contract.
c- bulges outwards when the Eustachean tube is blocked.
d- modifies the frequency of sound waves impinging on the ear.
Answer: A

187. The primary function of the middle ear bones is to:
a- protect the ear from loud sounds.
b- localize sounds
c- amplify sounds.
d- frequency discrimination.
Answer: C

188. Depolarization of the hair cells in the cochlea is caused by the influx of:
a- Na^{+} into the hair cells.
b- K^{+} into the hair cells.
c- Ca^{++} into the hair cells.
d- Cl^{-} into the hair cells.
Answer: B

189. If the patient is unable to hear low-frequency sounds, the damage to the basilar membrane is closest to:

a- oval window.
b- helicotrema.
c- modiolus.
d- spiral ganglion

Answer: B

190. The stereocilia of hair cells are embedded in the:

a- Reissner's membrane:
b- basilar membrane.
c- tectorial membrane.
d- rods of Corti.

Answer: C

191. Sour taste is elicited by:

a) alkaloids
b) amino acids
c) hydrogen ions
d) aldehydes

Answer: C

192. the electrical events in Rods are started when ……. Is formed:

a) Bathorhodopsin
b) Lumirhodopsin
c) Metarhodopsin I
d) Metarhodopsin II

Answer: D

193. The center of light reflex is:

a) Inferior colliculus
c) superior colliculus.
b) Pretectal nucleus
d) lateral geniculate body.

Answer: B

194. in Argyl-Robertson pupil:

a) light reflex is lost and near reflex is intact.
b) light reflex is intact and near reflex is lost.
c) Both light and near reflexes are lost.
d) Both light and near reflexes are intact.

Answer: A

195.the pupil is constricted during anesthesia in:

a- first stage b) Second stage c) third stage d) fourth stage.

Answer: C

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d) green is perceived when only green cones are stimulated.

Answer: A

162. In response to light, the rods hyperpolarize as a result of:

a) an increase in sodium pump activity of the inner segment.
b) a decrease in Na^+ conductance into the outer segment.
c) an increase in potassium conductance into the outer segment.
d) a decrease in sodium pump activity at the inner segment.

Answer: B

163. Lesion of the optic chiasma will lead to:

a) homonymous hemianopia.
b) bitemporal hemianopia.
c) blindness in both eyes.
d) Protanopia.

Answer: B

164. Regarding light adaptation, the following is likely to occur:

a) pupil size increases.
b) vitamin A converts into retinal.
c) rhodopsin concentration is very high.
d) photopigments in rods and cones are reduced into opsin and retinal.

Answer: D

165. In myopia:

a- the near point is farther than normal from the eye.
b- near vision is affected more than distance vision.
c- the eye tends to be longer than average from lens to retina.
d- a convex lens can correct the refractive error.

Answer: C

166. Constriction of the pupil increases:

a- refractive power of the eye.
b- spherical aberration.
c- depth of focus.
d- amount of light entering the eye.

Answer: C

167. During near vision, the following changes occur in the eye, EXCEPT:

a- the visual axes of the two eyes converge.
b- The depth of focus increases.